

Zoe Szymanski

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY **Expected May 2026**
Bachelor of Science, Mechanical Engineering, Minor in Aerospace Engineering
GPA: 3.43 | Deans list (Fall 2022, Spring 2025)
Current Coursework: System Dynamics, Introduction to Aeronautics, Mechanics of Materials, Fluid Mechanics
Portfolio: <https://sites.google.com/view/zoe-szymanski/project-page>

Governor's School for Science and Mathematics, Hartsville, SC **June 2022**
GPA: 5.31
National Honor Society, Student Ambassador, Head engineering tutor, Engineering lab assistant, Outreach and STEM programs student instructor

ENGINEERING EXPERIENCE

Baja SAE Racing Team, Cornell University, Ithaca, NY, *Subteam Lead* **Aug. 2022-Present**

- Work with a team of 50 students to build and race an off-road vehicle at the Baja SAE Collegiate Design Series.
- Rear bevel gear and casing designer.
- Currently designing the geometry of the 2024-2025 Drivetrain.
- Designed Rear Gearbox casing and Input shaft support structure.
- Achieved 13.4% weight savings on the gearbox casing.

REV Hardware Accelerator, Ithaca, NY, *Summer/Fall Intern/Spring intern* **May. 2024- Present**

- Worked with 21 teams on early prototyping and engineering support.
- Selected as the primary student engineer for Canoeist Carrier as part of the Protofacturing Accelerator.
- Created CAD models for prototypes and worked with founders on DFM plans.

CMMC Machine Inc., Charleston, SC, *Summer Intern* **Jun. 2023- Aug 2023**

- Engineering team member for stress-relieving processes for USN submarine shafts.
- 3D-modeled existing submarine drivelines for CNC repair processes.
- QA team member focusing on cuts, welds, and final tolerance inspections.

RESEARCH EXPERIENCE

Clemson University SPARK Charleston, SC, *Student researcher*

- Designed and manufactured a gimbal and pendulum setup to model thrust vector control.
- Created a PID controller to create a control loop that achieves critical damping.

LEADERSHIP EXPERIENCE

Cornell Baja Geartrain Subteam Lead, Cornell University **Aug. 2024-Aug. 2025**

- 2024-2025 Geartrain lead.

Computer-Aided Manufacture MAE 5250, Cornell University, TA, Lab Assistant **Aug. 2024-Present**

- Advanced machining course focusing on the use of the HAAS machines.
- Assists in lecture presentations and lab sections.

Emerson Shop Supervisor, Cornell University **Jan. 2023-Present**

- Supervisor for student project team machine shop.

Space Force Reserve Officer Training Corp, Cornell University, *Cadet* **Aug. 2022-Present**

- Public Affairs Squadron Commander.
- Cadet Captain Space Force ROTC.

SPECIALIZED SKILLS

Programming Languages: MATLAB, Python
Programs: SolidWorks, Siemens NX, Fusion, CAM, KISSSOFT, Ansys
Fabrication Skills: Operation of CNC lathes and HAAS mills (Blue Apron machinist at Cornell), Waterjet, 3D printing